

Ravi Narayan Mahawar

Applied AI & ML Engineer / Product Thinker

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SUMMARY

Applied AI and ML engineer with over five years shipping production systems. Has taken LLM-powered features from an empty repo to real users in a regulated insurance domain, and owned them once they were live: an agentic policy-bind evaluation engine built on LangGraph and RAG, an MCP server serving internal agents, and multi-agent Claude pipelines that simulate and regress on hardened tasks. Works across the stack in Python, Golang, Java and React on AWS, with a research background in reinforcement learning and multi-agent RL where inventing an approach and proving it worked was the job.

SKILLS

Applied AI: LLM applications, RAG (retrieval-augmented generation), LangGraph, Agentic workflows, MCP (Model Context Protocol), Tool calling, Claude agent orchestration, Multi-agent systems, Agent simulation and regression, Task hardening, AI-assisted development

ML & Research: PyTorch, Keras, Scikit-Learn, Reinforcement learning, MARL, Quantum computing, MPI, OpenMP, OSINT

Languages: Python, Java, Golang, JavaScript, TypeScript, Ruby, C/C++

Backend & APIs: Microservices, REST, GraphQL, Node/Express, Ruby on Rails, Celery, Kafka, RabbitMQ, Caching, Schema design, Monitoring and dashboards

Cloud & Infrastructure: AWS, AWS Lambda, AWS S3, EC2, ECS/EKS, RDS, CloudWatch, SNS, Docker, Kubernetes, Git, GitHub, CI/CD, Logging and monitoring

Databases & Storage: PostgreSQL, MySQL, MariaDB, SQLite, MongoDB, Cassandra, DynamoDB, Redis, Memcached

Analytics & Big Data: Snowflake, Elasticsearch, Presto/Trino

Frontend: TypeScript, React.js, Next.js, Angular, Vue.js, Redux, Zustand, Context API, Material-UI, Tailwind CSS, Bootstrap, Chart.js, D3.js, Recharts, Axios, SWR, React Query

Testing: Jest, React Testing Library, Cypress, Pytest

EXPERIENCE

Senior Full Stack Developer at MyChoice Financial Inc.

Toronto, ON | 01/2026 to Present

- Shipped Broker After-Hour Referral, letting brokers get Managing Partner sign-off outside business hours to fast-track binds instead of losing the sale to the appeal process. Drove 1,000+ referrals in two weeks and CAD 6M+ in business retained in the first two months.
- Built a runtime policy-bind evaluation engine, an LLM agentic workflow in LangGraph grounded in the underwriting guidelines through RAG, that flags plausible issues at bind time so brokers correct errors before exposure reaches audit.
- Took the MCP (Model Context Protocol) server and the ChatGPT-native quoting plugin to production, giving internal agents governed access to company data and letting customers get preliminary quotes inside an AI assistant.
- Added the CI/CD pipeline behind those services, and owned the broker-facing product area across successive releases, iterating with broker and Managing Partner feedback toward product-market fit.
- Started features from ambiguous problem statements rather than specs, refining scope through regular feedback and expectation rounds with brokers and Managing Partners before and during the build.
- Engineered a policy auditing platform running 111+ validations across underwriting, soft-fraud and premium-leakage to surface pricing exposure on bound policies.
- Led a caching layer precomputing the Sales, Service and Book of Business dashboards into mathematically derived buckets, cutting load times from seconds to under one second with separate admin and broker views.

- Redesigned the homepage around modular components to host new broker features and give Managing Partners and admins a broker's-eye view of the platform, increasing visibility into broker activity.
- Built a real-time sales performance and contest dashboard for brokers and leadership to track individual and branch-level metrics, with reliable monthly leaderboard rollovers and no manual intervention.
- Optimized customer-facing flows and document fetching systems to reduce platform latency, improving broker throughput and application completion rates.
- Drove CAD 2M+ in retained business in the first two weeks of the Broker After-Hour Referral launch (July 2026).

Founder / AI & ML Research Consultant at Dashwood ThinkTank

Toronto, ON | 04/2026 to Present

- Founded an independent AI/ML research consultancy taking on applied research engagements for clients under NDA.
- Researched task hardening for agents, training them on difficult problems with reinforcement learning and other applied ML methods.
- Designed multi-agent Claude pipelines that simulate and regress on hardened tasks, measuring them against target outcomes and fail rates.
- That pipeline became the client's foundational multi-agent swarm workflow, taken up by other contributors and adopted across all of their projects.

Research Partner at Research ThinkTank

Toronto, ON | 09/2025 to Present

- Co-founded a multidisciplinary non-profit research lab with Dr. Shohini Ghose and Dr. Toorjo Ghose.
- Research on the NLSY mental health dataset, a US government-regulated longitudinal survey, alongside combinatorial mathematics and photon light activity correlation, spanning quantum computing, AI/ML and high-performance computing.
- Research directions intersect mathematics, quantum computing, AI/ML and high-performance computing.
- Collaborators include Dr. Ilias Kotsireas and Dr. Nirosha Murugan.
- Models and datasets are held private under IP protection pending publication of the associated papers.

ML Engineer at Project Ploughshares

Kitchener, ON | 06/2025 to 09/2025

- Developed an open-source intelligence (OSINT) monitoring system to aid in identifying potential arms embargo violations and support global disarmament efforts.
- Implemented a global web scraper using Crawl4AI that performs shallow digs for relevant data, escalates to deeper exploration on human approval, and uses LLMs to summarize findings and surface geolocated resources worldwide.
- Made new resources discoverable for the team while keeping a human in the loop for source reliability, eliminating manual repetitive work across data ingestion, context mapping and association.

Quantum Computing & ML Research Assistant at Wilfrid Laurier University

Waterloo, ON | 09/2022 to 08/2025

- Developed optimized solutions for D-optimal, Legendre Pair and Hadamard Matrix problems using serial and MPI implementations, testing performance prior to quantum resource implementation.
- Designed reinforcement learning algorithms that solved combinatorial problems at two-digit problem sizes, then reached three-digit sizes by transferring the learned policies rather than searching from scratch.
- Engineered three quantum algorithms with D-Wave for mathematical problems, focusing on performance optimization and resource efficiency.
- Built an explainable MARL framework on top of VIPER, evaluating performance across 3 games and 40+ parameter combinations on AWS servers.

Senior Software Engineer at Multiplier

Remote | 05/2021 to 08/2022

- Improved customer onboarding efficiency by 60% by developing a dynamic salary breakdown service for 40+ countries and currencies.

- Increased platform communication capacity, handling 100+ email notifications daily, by designing a scalable cross-service email solution.
- Reduced processing latency by 40% by integrating Amazon MSK and establishing 10+ asynchronous channels between microservices.
- Accelerated contract workflows by 75% by automating document generation across five microservices through PandaDoc integration.
- Mentored junior engineers and led the extraction of microservices from the existing monolith, building the new services from scratch.

Senior Software Engineer at MakeMyTrip

Bangalore, India | 06/2019 to 05/2021

- Led the project to calculate and apply day-wise offer combinations producing the cheapest booking for end users across platforms.
- Implemented dynamic programming algorithms for promotion calculations, enhancing offer application logic on multiple booking platforms.
- Drove a 20% increase in bookings, adding 500 room nights daily, by leading the Day Wise Promotion project.
- Spearheaded adoption of closed user group discounts, achieving CUG application in 50% of all bookings through models, API integration and business logic.
- Migrated three TIER-1 services from ECS to EKS.

Software Engineer at Piggy

Bangalore, India | 08/2018 to 06/2019

- Increased API performance by 40% across 10+ user-facing endpoints.
- Developed and deployed a Celery system with auto-scaling capacity, handling 150+ parallel tasks per day.
- Achieved 99.99% uptime for payment services by integrating Razorpay and automating all refund processes.
- Mentored 3 SDE interns and led development of 3 key features including Simply Save, Digital KYC and One-Link Invest.

SDE Intern at Amazon

Bangalore, India | 01/2018 to 06/2018

- Independently developed a caching mechanism that periodically refreshes the latest version of all resources, performs schema validation and maintains an in-memory cache, eliminating a database hit on every request.
- Independently developed an S3-based data source that periodically refreshes data from the service.
- Re-architected the service client library to fall back to the S3 source when the service is unavailable.

PROJECTS

APP-FractalProject, C/C++, MPI, OpenMP

- Generates fractal images using parallel processing and converts the raw binary output into a PNG image.
github.com/raviwaar/APP-FractalProject

SLAM (Smart Lab Availability Manager), Python, OneDrive API

- Edits an Excel file in OneDrive to find available lab slots and book them seamlessly. github.com/raviwaar/SLAM

911 Calls Analysis Platform, Spring Boot, Java, Jupyter

- Spring Boot web app integrating with Jupyter Notebooks to analyze and visualize 911 call data.
github.com/raviwaar/AI-911-Calls-Platform

React Job Portal, React, JavaScript

- Job portal with flows for both job seekers and employers. github.com/raviwaar/react_job_portal

EDUCATION

Master of Applied Computing, Wilfrid Laurier University

Waterloo, Canada | 09/2022 to 08/2025

B.Tech., Computer Science, College of Engineering and Technology

Bhubaneswar, India | 08/2014 to 05/2018